



Al-Balqa' Applied University

Faculty of Engineering Technology, Department of Basic Scientific Sciences

Final Exam - Linear Algebra I

Date: 14/6/2022

Part A: Multiple choice

(24 marks)

-Be sure that you have 11 different questions.

-Complete the following table by putting (the correct answer letter) in the table.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
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Q1) Suppose that the augmented matrix for a system of linear equations has been reduced by row operation to the given form:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 2 & 4 \end{array} \right]$$

The solution of the linear system (respectively x, y, z) is

- a) $2, -3, 2$ b) $2, -1, 1$ c) $-1, 5, -1$ d) $4, -3, 2$

Q2) The values of k , if any that satisfy the equation $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ is

- a) 2 b) 1 c) 0 d) -1

Q3) If $(I - A^T)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix}$, then $\text{Tr}(A) =$

- a) 4 b) 3 c) -4 d) 2

Q4) For some nonsingular square matrix A , assume that $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -1 & 2 \end{bmatrix}$ the solution of the system

$$AX = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \text{ is } \quad \text{a) } X = \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix} \quad \text{b) } X = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{c) } X = \begin{bmatrix} -1 \\ 1 \\ -4 \end{bmatrix} \quad \text{d) } X = \begin{bmatrix} -1 \\ 0 \\ 4 \end{bmatrix}$$

Q5) The system $\begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

- a) no solution b) two solution c) infinitely many solutions d) unique solution

Q6) The value of k for which $\det(A) = 0$, where $A = \begin{bmatrix} 7 & 0 & 0 \\ -3 & k-6 & 0 \\ 0 & 0 & k-4 \end{bmatrix}$ is

- a) 4, 3 b) 1, 6 c) 4, 6 d) 1, 3

Q7) If $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = 4$ evaluate the $\begin{vmatrix} a & b & c \\ -2g & -2h & -2i \\ d & e & f \end{vmatrix} =$

- a) -8 b) 12 c) -12 d) 8

Q8) The value of x , if any that satisfy the equation $\begin{vmatrix} 2 & x+2 \\ -1 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 8 \\ -2 & x-2 \end{vmatrix}$ is. a) 12 b) 4 c) 2 d) 6

Q9) Let the vectors $v = (2, 7)$, $w = (-4, 8)$ the components of the $v+2w$ is
a) (6, 10) b) (16, 10) c) (6, -10) d) (-6, 17)

Q10) The terminal point of the vector that is equivalent to $u = (3, 4, 6)$ and whose initial point is $B(-3, -1, 2)$,
a) (-1, 3, 8) b) (8, -1, 1) c) (2, -1, 8) d) (-1, 3, 4)

Q11) Let $u = (2, 3, 0)$ and $v = (-1, 2, 2)$ the distance between u and v is a) $\sqrt{17}$ b) $\sqrt{14}$ c) $\sqrt{11}$ d) $\sqrt{10}$

Q12) Let $A = \begin{bmatrix} 2 & 0 \\ 4 & -3 \end{bmatrix}$ the Eigen values = a) 2, 3 b) -2, 3 c) -2, -3 d) 2, -3

Part B: Fill the blank

(12 marks)

Q1: Let $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ Then $\|(2A)^{-1}\| = \underline{\hspace{2cm}}$

Q2: Let $A = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$ the $\text{adj } A^{-1} = \underline{\hspace{2cm}}$

Q3: Let $V = \mathbb{R}^2$ define the operations $(x_1, y_1) \oplus (x_2, y_2) = (x_1 + 2x_2, y_1 - 2)$ and $c \odot (x, y) = (cx, c+y)$ then
 $4 \odot ((2, 3) \oplus (-2, 5)) = \underline{\hspace{2cm}}$

Q4: Let $u = (0, 1, -1)$, $v = (1, 2, 0)$ then the cosine of the angle θ between u and v is $\underline{\hspace{2cm}}$

Q5: Let the vectors $u = (2, k)$ and $v = (3, 6)$, the value of k such that u and v are orthogonal $\underline{\hspace{2cm}}$

Part C: Written questions

(12 marks)

Question 1: Let $A = \begin{bmatrix} 1 & 5 & 6 & 2 \\ 2 & 4 & -2 & 1 \\ 6 & 12 & -6 & 3 \\ 1 & 0 & -1 & 2 \end{bmatrix}$ show that $\det(A) = 0$, without directly evaluating the determinant, by using

the property which we

Question 2: let $W = \{f \in p_3 : a_0 = 0\}$ where p_3 set of vectors for 2^{nd} order polynomials. Show that W are subspace of the vector space.

Question 3: Let $p_1 = 1-x$, $p_2 = 3+3x-2x^2$, $p_3 = 1+3x-x^2$ are p_1, p_2, p_3 linearly dependent or independent? Why.

Question 4: Let $\det(A) = 2$, $\det(B) = 4$ find $\det(A^{-1}B)$.

With my best wishes .

$$\text{adj } A^{-1} = \frac{A}{\det A}$$

$$\text{adj } A^{-1}$$

$$A \times \text{adj } A^{-1} = I$$

Q1) Can the vectors $x = (2, 3, -1)$, $y = (-4, 1, 4)$ be components of the $x = 2y + z$ in $\mathbb{R}^3$ (10 marks)

Q2) The eigenspace of the matrix A is equivalent to $z = (2, 4, 6)$ and whose initial point is $B = (-5, 1, 7)$.
 $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ (10 marks)

Q3) Find the unit vector that has the same direction as $\vec{v} = (1, 2, 3)$.
 $\vec{u} = \frac{1}{\sqrt{14}}(1, 2, 3)$ (10 marks)

Q4) Let $u = (2, 1)$ and $v = (-3, 2)$ find $\text{proj}_u v$ and $\text{proj}_v u$.
 $\text{proj}_u v = \frac{v \cdot u}{u \cdot u} u = \frac{-6+2}{5} (2, 1) = (-4/5, -2/5)$
 $\text{proj}_v u = \frac{u \cdot v}{v \cdot v} v = \frac{-6+2}{5} (-3, 2) = (6/5, -4/5)$ (10 marks)

Q5) Let the vector $u = (2, 1, -1)$ and $v = (1, 0, 1)$ find $\text{proj}_u v$.
 $\text{proj}_u v = \frac{v \cdot u}{u \cdot u} u = \frac{2+0-1}{3} (2, 1, -1) = (1/3, 1/3, -1/3)$ (10 marks)

Q6) Let $A = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$ the eigen values are $\lambda_1 = 2, \lambda_2 = -1$.
 Part B: Find the matrix P such that $P^{-1}AP = D$.
 $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ (10 marks)

$$A^{-1}$$

Q7) Find the diagonal matrix D such that $A = PDP^{-1}$ for $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.
 $D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ (10 marks)

Q8) Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ find A^2 .
 $A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{pmatrix}$ (10 marks)

$$A^+$$

Q9) Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ find A^{-1} .
 $A^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ (10 marks)

Q10) Let $u = (1, 2, 3)$ and $v = (-3, 4, 2)$ find $\text{proj}_u v$.
 $\text{proj}_u v = \frac{v \cdot u}{u \cdot u} u = \frac{-3+8+6}{14} (1, 2, 3) = (1, 2, 3)$ (10 marks)

Part C: Written questions

Question 1: Let $u = (1, 2, 3)$ and $v = (-3, 4, 2)$ find $\text{proj}_u v$.
 $\text{proj}_u v = (1, 2, 3)$ (10 marks)

Question 2: Let $W = \{ (x, y, z) \in \mathbb{R}^3 : x + y + z = 0 \}$ find $\dim W$.
 $\dim W = 2$ (10 marks)

Question 3: Let $P_1 = 1+x, P_2 = 1+x^2, P_3 = 1+x^3$ are P_1, P_2, P_3 linearly independent or not? Why?
 They are linearly independent. (10 marks)

Question 4: Let $\det(A) = 2, \det(B) = 3$ find $\det(A^T B)$.
 $\det(A^T B) = 2 \cdot 3 = 6$ (10 marks)